FORMULABENCH

FormulaBench v2

Guarded workbook execution for Qwen3.8-27B

Research: Excel Formula Generation
Encode x Ylookup Rebuild Private Markets Hackathon

TYPED PLAN


```
{
  "route": "operations",
  "edits": [
    {
      "sheet": "Returns",
      "cell": "H9"
    }
  ]
}
```


ILLUSTRATIVE STRUCTURE

CURRENT DEFAULT CANDIDATE

UNSCORED AT CAPTURE

6 September 2026

Three records answer three different questions

FormulaBench v2 keeps execution evidence separate from benchmark accuracy.

ENGINE

Can the path run?

Typed operations and screened Python execute under bounded checks, then save to a fresh copy.

Code, tests and route canaries

MIGRATION PARITY

Did the wrapper preserve behaviour?

A credential-free replay compared retained ExactSource outcomes with the FormulaBench v2 wrapper.

400 checked, zero mismatches

BENCHMARK

Are the answers correct?

Only a complete output scored by the unchanged organiser evaluator can establish v2 accuracy.

Fresh v2 result not yet published

These records are reported separately. None is converted into a FormulaBench v2 score.

One model, two bounded execution routes

Qwen/Qwen3.8-27B produces a typed plan from bounded workbook context.

QWEN/QWEN3.8-27B

Initial plan, then at most one repair or truncation recovery

CONCURRENCY 4

TARGET CELLS

Typed workbook operations

Cell-level tasks can use only the declarative operation route. Every edit carries an explicit sheet and address.

Containment and formula checks before write

LARGER SHEET TASKS

Typed operations or screened Python

Sheet-level transformations can use the same operations or screened Python with limited imports and resources.

Temporary execution before publication

The second-call allowance is exclusive. A task never gets both repair and truncation recovery.

A valid workbook can still contain a wrong answer

V2 contains failures it can verify. The organiser evaluator still decides correctness.

01

Context

The needed sheet, range or existing formula is missing from the bounded context.

02

Coverage

The plan omits a target, duplicates an edit or writes outside the allowed area.

03

Workbook

The plan violates formula policy, resource limits or the save and reopen contract.

04

Semantics

The output opens and recalculates, yet the investment result is still wrong.

Execution checks can reject unsafe work. They cannot certify a financially correct answer.

V2 publishes only after the complete guarded path

The pristine workbook remains the recovery source throughout the task.

01
Load
Pristine input

02
Context
Bounded evidence

03
Plan
Initial call

04
Gate
Parse and check

05
Recover
Optional second call

06
Execute
Temporary copy

07
Publish
Validated output

PASS PATH

Validate targets, formulas, resources and workbook save. Then publish atomically.

REJECT PATH

Discard the work and copy the pristine input to the submission output.

Checks before a workbook can be published

Typed operations and the sheet route share a fail-closed publication boundary.

Boundary	Required evidence	On failure
Target containment	Every edit remains inside the declared worksheet and target area.	**Reject the plan**
Resource policy	The selected route stays within bounded time, memory and import limits.	**Stop execution**
Formula policy	Formula assignments pass the workbook formula checks before publication.	**Reject the plan**
Workbook save	A fresh copy saves successfully and reopens under the output contract.	**Discard the copy**
Output integrity	The published workbook and recorded prediction agree on the task outcome.	**Use pristine fallback**

Rejected plans preserve the pristine workbook.

Migration parity reproduced all 400 retained outcomes

A credential-free replay compared the FormulaBench v2 wrapper with retained ExactSource artefacts.

400
tasks checked

369
accepted plan replays

31
pristine fallback checks

0
wrapper mismatches

88.13 seconds
host wall time, including container startup

Accepted plans

Rebuilt workbook content and compared the OOXML member inventory and payloads after normalising core timestamps.

Fallback plans

Required byte-identical output against each pristine input workbook.

Zero model calls. No golden workbooks read. This is migration parity, not benchmark accuracy.

The evaluator grades results, not plausible output

A complete v2 score requires the unchanged organiser evaluation path.

Stage	Input	What it establishes
1 Recalculate	Generated workbook opened by LibreOffice	Formulas produce current workbook values
2 Compare	Every declared target cell	Exact cell matches and mismatches
3 Grade task	Complete target set	A pass only when the whole task is correct
4 Aggregate	All 400 evaluated tasks	Task pass rate and cell accuracy

Opening, saving and recalculating are necessary checks. They are not a substitute for target-cell correctness.

Two selected canaries exercised both execution routes

Each case ran through LibreOffice and the unchanged organiser evaluator.

TYPED OPERATIONS

Task 54513

1 / 1

target cell correct

Cell-level manipulation
One model call
Runtime status: ok

SCREENED PYTHON

Task 23-24

5,510 / 5,510

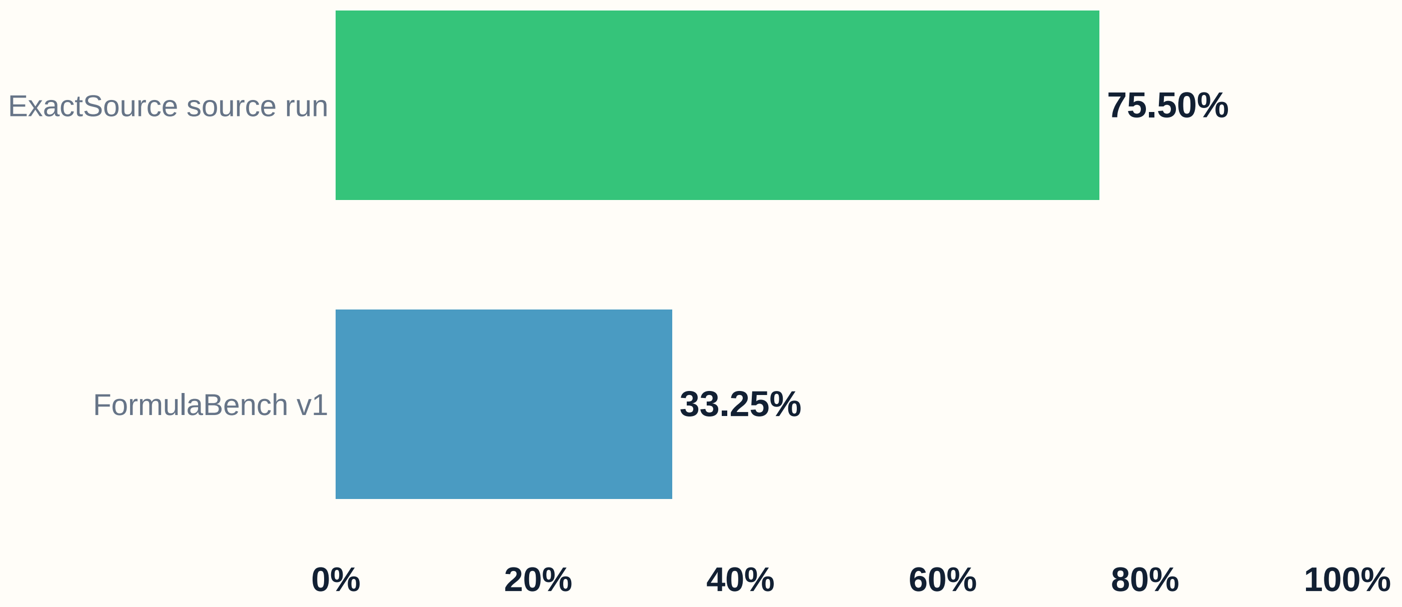
target cells correct

Sheet-level manipulation
One model call
Runtime status: ok

These are selected route diagnostics. A two-case pass count cannot estimate performance on the 400-task benchmark.

Historical scores stay attached to their recorded runs

No earlier result is transferred to FormulaBench v2.



FORMULABENCH V2

UNSCORED

No complete 400-task result had been published at deck capture.

Recorded run	Task result	Cell result
FormulaBench v1	133 / 400 33.25%	40.56%
ExactSource source run	302 / 400 75.50%	80.06%

Run it, inspect it, then score the complete output

The public repository keeps the runtime, parity record and historical score distinct.

RUN V2

```
uv run python data/download.py
./scripts/run_docker.sh data/spreadsheetbench_verified_400 submissions/formulabench-v2
```

Set TINKER_API_KEY privately in the shell before a paid run.

EVIDENCE ANCHORS

experiments/v2_migration_validation.json
submissions/formulabench/results.json
SUBMISSION.md

CAPTURE STATUS

Current default

FormulaBench v2

Benchmark state

Unscored at capture

6 September 2026

PUBLIC REPOSITORY

github.com/MasteraSnackin/FormulaBench

Private benchmark remains unseen